

DATA DRIVEN METHODS TO ANALYZE WAVE BUOY OBSERVATIONS

A R Kambekar¹, M C Deo², Mansi Daga³, Suhasini Sakhare⁴

^{1, 2, 2, 4} Indian Institute of Technology, Bombay, Mumbai India 400 076

¹ e-mail: arkambekar@iitb.ac.in, ² e-mail: mcdeo@civil.iitb.ac.in,

³ e-mail: mansi_daga@iitb.ac.in, ⁴ e-mail: suhasini_313@iitb.ac.in

Abstract:

Recent data driven methods such as genetic programming (GP), model tree (MT), support vector regression (SVR), and locally weighted projection regression (LWPR) look attractive in carrying out a variety of tasks in coastal and ocean engineering. This paper provides an exposure to their applications made by authors to hindcast significant wave heights, to estimate wave surface spectral density and to inversely evaluate wind from waves. The field measurement used for this purpose mainly belonged to two selected locations in Arabian Sea along the west coast of India. The studies indicated that the various data driven methods can be effectively used to evaluate causal, temporal as well as inverse relationships between various wave and wind parameters and also for resolution of average wave parameters into a set of spectral densities. It is expected that the studies reported in this paper would pave the way for solutions to more complex and difficult tasks in coastal and ocean engineering such as estimation of uncertain design parameters and spatio-temporal evaluation of ocean environment.

Keywords: locally weighted regression, genetic programming, support vector regression, model trees

INTRODUCTION

The problem solving in ocean engineering has been traditionally dominated by model-driven approaches that either conceptualize the physical process into mathematical equations and solve these equations using analytical, empirical or numerical schemes, or assume some dependency structure between the known and unknown quantities beforehand, without going in for the process knowledge, and evaluate the same with the help of measured data. However with the advent of computer technology various data driven methods are becoming available for their applications in this field of engineering and since last decade artificial intelligence or soft computing methods such as artificial neural networks (ANN), fuzzy logic and genetic algorithm have found their way in tasks like, damage detection of ocean structures and wave forecasting. (See for example Jain and Deo, 2006, for a review

on ANN applications.) Of more recent origin among the data driven approaches are the techniques of genetic programming (GP), model tree (MT), support vector regression (SVR) and locally weighted projection regression (LWPR), for which sizable number of applications in the sea water are yet to be reported. (See Appendix I for details of these techniques as well as the literature survey). This paper tries to fill up this void to some extent. It illustrates the usefulness of these tools to carry out certain common tasks in the sea water such as wave hindcasting, wave spectrum evaluation from average wave parameters and also rather uncommon works such as deriving wind information from measured waves.

WAVE HINDCASTING

Floating wave rider buoys denote the most common type of in-situ wave data collection device. Such buoys float over the wave surface, undergo vertical accelerations as a result and by double integrating these produce instantaneous sea surface displacement values. A buoy typically yields one value each of significant wave height, H_s , and average wave period, T_z , every one or three hours.

Three-hourly measurements of these parameters along with those of wind speed, U and direction θ at a location code named 'DS1' off 'Goa' along the west coast of India (Figure 1) were considered in this study. The observations were made by National Institute of Ocean Technology located at Chennai in India. The period of data collection varied from 1998 to 2006.

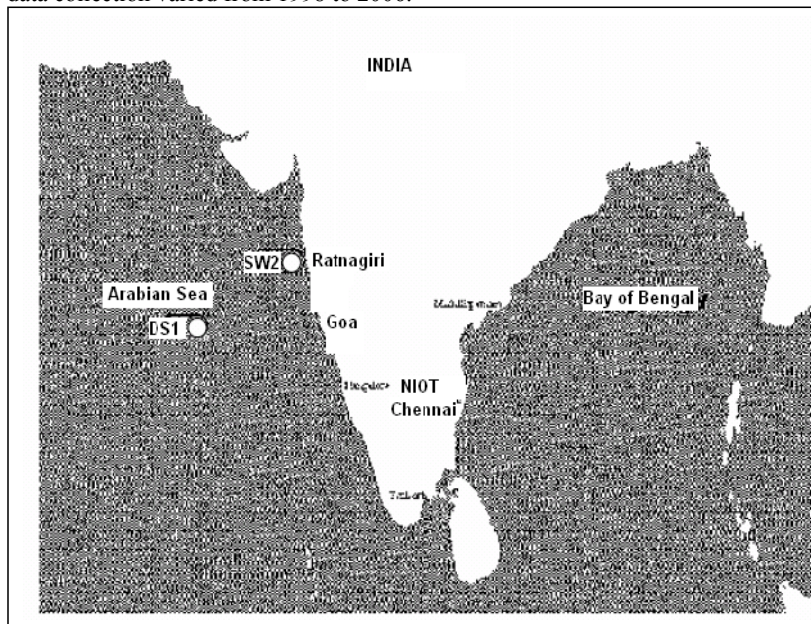


Figure 1. Indian coastline showing stations: DS1, SW2

(<http://www.niot.res.in/ndbp>)

Site-specific evaluation of waves from wind parameters is traditionally done through an empirical approach such as the coastal engineering manual method (U.S. Army, 2003). Recently the data driven approach of ANN has been used; as in Deo et al. (2001) and Zamani et. al. (2008). In this work we have employed GP as well as MT for this purpose in order to see if how these techniques perform in such causal mapping. Out of these two approaches the GP is a purely non-linear method while the MT is a piecewise linear model. The GP consists of generating innumerable combinations of variables and mathematical operators based on genetic process simulations and selecting the fittest combination in the end, while the MT divides the input space into sub-spaces and fits a linear model in each one. Appendix I gives some more conceptual information on GP and MT.

In this study ‘Hs’ and ‘Tz’ values were derived at the current time step based on the causative parameters of wind speed and wind direction. The generation of waves from wind may depend not only on wind characteristics of the current time but also of the preceding time steps and hence a sequence of ‘U’ and ‘ θ ’ over preceding 24 hours was used to arrive at most acceptable values of the wave height and period. Various trials were made by varying this input time sequence from 24 hr to 0 hr (current time). The GP and MT models were calibrated using data from 1998 to 2002, while they were tested with the help of measurements that ranged from 2002 to 2006. Both the sets involved severe monsoon and relatively calmer non-monsoon conditions and had comparable range of means, standard deviations and minimum and maximum values. Over the whole set of data the significant wave height had a mean of 1.72 m and standard deviation of 1.17 m. The highest ‘Hs’ was 7.81 m. The same values for ‘Tz’ were 5.63 sec, 1.23 sec and 10.66 sec, respectively.

Tables 1 and 2 show the performance of both GP and MT models respectively in estimating Hs values in terms of the error statistics of correlation coefficient R, mean absolute error, MAE and root mean square error, RMSE in between the predicted and the observed values over the testing set of data. Figures 2 (a) and (b) show as examples how the simulated and the actual values of ‘Hs’ corresponded with each other qualitatively in case of MT and GP, respectively.

Table 1. Estimation of Hs at DS1 with GP

No.	Input	R	MAE (m)	RMSE (m)
1	U, θ (at current time step)	0.90	0.11	0.55
2	U, θ (over preceding 24 hr)	0.93	0.10	0.47

Table 2. Estimation of Hs at DS1 with MT

No.	Input	R	MAE (m)	RMSE (m)
1	U, θ (at current time step)	0.90	0.37	0.54
2	U, θ (over preceding 24 hr)	0.92	0.33	0.50

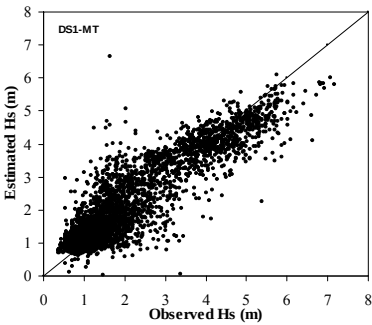


Figure 2(a) MT-estimated versus observed Hs (R = 0.92)

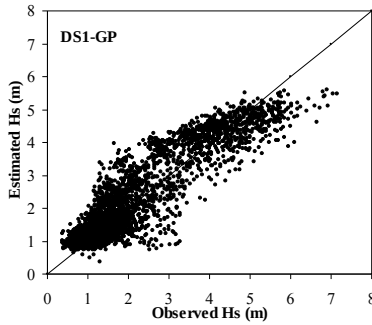


Figure 2(b) GP-estimated versus observed Hs (R = 0.93)

It may be seen from these Tables that although accounting for the preceding 24 hour's wind speed and direction helps the calibration process, comparable accuracy can still be achieved if the hindcasting is done with the help of data of only the current step. Figure 2 also shows that both the methods, GP and MT, perform satisfactorily in the given task of wind-wave simulation as reflected in high values of R, and low values of MAE, and RMSE (Table 1, 2). Although the magnitudes of the error statistics do not indicate a significant difference in between the performance of GP and MT, the qualitative scatter diagram and time histories show the tendency of MT to estimate the higher waves more correctly. This is noteworthy since MT is not a purely non-linear model like GP and one would have expected GP to do well in all ranges of waves. However it appears that the splitting of input domain into sub-domains helps in better simulating all ranges of waves including the higher ones.

The performance of GP and MT models in the estimation of average zero cross period using the input of wind speed and direction of the current time step and that of preceding 24 hr. is shown quantitatively in Tables 3 and 4 and qualitatively in Figure 3. This leads to the same inference as in case of 'Hs' indicated in the preceding paragraph. In general the estimation of 'Tz' is less satisfactory than that of 'Hs' and this can only be expected in view of the highly varying nature of 'Tz' values.

Table 3. Estimation of Tz at DS1 with GP

No.	Input	R	MAE (sec)	RMSE (sec)
1	U, θ (at current time step)	0.75	0.07	0.86
2	U, θ (over preceding 24 hr)	0.76	0.05	0.85

Table 4. Estimation of Tz at DS1 with MT

No.	Input	R	MAE (sec)	RMSE (sec)
1	U, θ (at current time step)	0.76	0.64	0.84

2 U, θ (over preceding 24 hr) 0.77 0.63 0.84

ESTIMATION OF WAVE SURFACE SPECTRAL DENSITY

A wave spectrum represents distribution of wave energy against various wave frequencies. The traditional method of estimating the spectrum is based on using the given values of significant wave height and average wave period to evaluate empirical spectral relationship such as Pierson-Muskowitz and JONSWAP (Sarpkaya and Isacson, 1981). Past studies have reported deficiency in this approach in global applications and hence Naithani and Deo (2004), and, Namekar and Deo (2005) used ANN for this purpose. Considering the uncertainty of the underlying problem and in view of the success of ANN the use of support vector regression was now attempted by us to evaluate sea surface spectral density values for given ‘Hs’ and ‘Tz’ parameters. The basic idea of SVR is explained in Appendix I. As can be noticed therein the SVR essentially linearizes the non-linear input space through Kernel based transformations (rather than only hard splits as in MT) and fits linear multiple regressions in each sub-space. It thus essentially targets high dimensionality of the problem and hence appears suitable for the application under consideration.

The SVR models were developed to estimate the wave surface spectral density over a wide range of wave frequencies from average wave parameters of significant wave height and average zero cross period. The model of SVR was built on the training set of data and was validated for the test samples. The measurements available on the web site: http://www.ndbc.noaa.gov/station_page.php?station=44008 and belonging to a period of 5 years and ranging from 1999 to 2003 at station: 44008 in the Gulf of Maine, USA were considered. For this site hourly measurements of spectral density functions along with corresponding frequencies, significant wave height and average wave period were extracted. In total 550 short term sea states were randomly selected, out of which 500 were used for training and 50 were employed for testing. It was found necessary to develop two separate models for higher and lower sea states in order to impart proper training. The low state corresponded roughly to ‘Hs’ less than 1.5 m. This value was empirically arrived at by trials.

In general the SVR was able to satisfactorily derive the spectral densities as seen in Figure 3 a and b as examples and as reflected in high values of the correlation coefficient, R, shown in these figures for a variety of spectral shapes. Figure 3 a and b show examples of how the SVR yielded spectra compare with the actual observed ones for the both stations. The values of the correlation coefficients noticed ranged mostly from 0.7 to 0.9, although occasionally they touched low magnitudes of 0.5, while those of MAE and RMSE varied from 0.10 m²/Hz to 10.06 m²/Hz and 0.01 m²/Hz to 3.06 m²/Hz for various sea states.

It is clear from the results that the SVR model was able to predict fairly satisfactorily the shape of wave spectrum from the short term average parameters of H_s and T_z. However the exact peak density was found to be generally underestimated or overestimated, like most theoretical spectra.

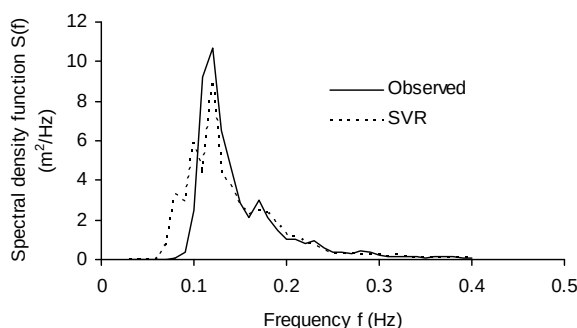


Figure 3 (a). Comparison of wave spectra at 21 Hr on 24/07/2003 (R=0.85)

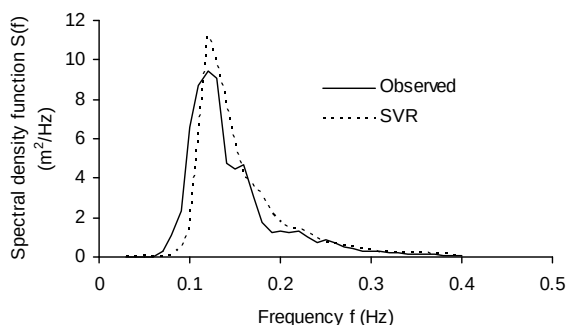


Figure 3 (b). Comparison of wave spectra at 16 Hr on 16/12/2003 (R=0.91)

ESTIMATION OF WAVES FROM WIND INVERSELY

As mentioned earlier, the wave rider buoys measure sea surface elevations that yield 'Hs' and 'Tz' values every one or three hour with the help of vertical accelerometers fitted on board. They also record wind speed and its direction thorough anemometers located at a considerable height above the mean sea level. Sometimes it is found that wave data continue to get collected, but not the wind data, due to the hardware failure in which the wind anemometer situated at a high location from the sea level gets disconnected from rest of the floating platform. This happens especially in storms due to heavy bending moments at the base of the connection. A need was therefore felt to retrieve the missing wind speed from the available wave information. The derivation of wind from waves is a problem of inverse modeling and needs to be solved by data driven methods since physical process modeling is not possible in this case, and further the underlying relationship is highly uncertain and hazy. We applied LWPR along with GP and MT to do so. The LWPR, like SVR, categorizes the input domain into sub-domains for linearity by taking projections of data onto different orthogonal directions and fits linear models in each and thereafter takes a weighted average of these. (Appendix I). The core LWPR code used in this study was taken from Toussaint (2006). The observations at location: SW2 (Figure 1) made by

NIOT India in Arabian Sea used in this work were available from January'05 to July'06 out of which training data belonged to the period: January'05 to January'06 and

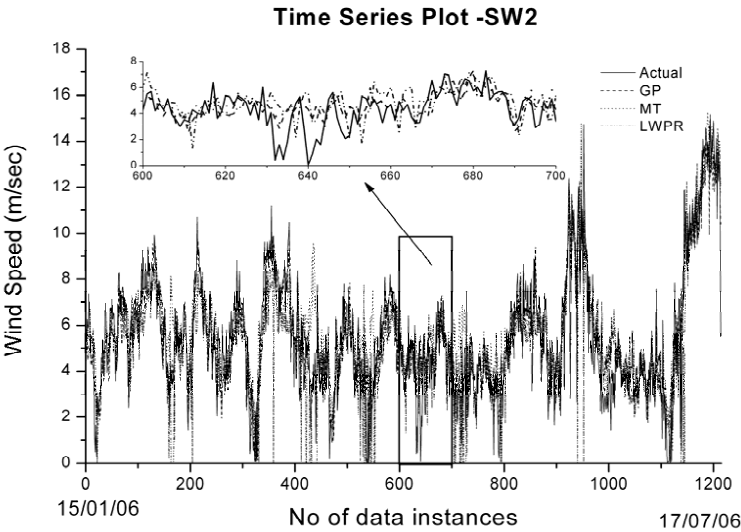
Table 5. The comparison of error measures during testing

Error	Methods		
	LWPR	GP	MT
R	0.81	0.91	0.84
RMSE (m/s)	1.68	1.13	1.47
MAE (m/s)	1.20	0.85	1.10

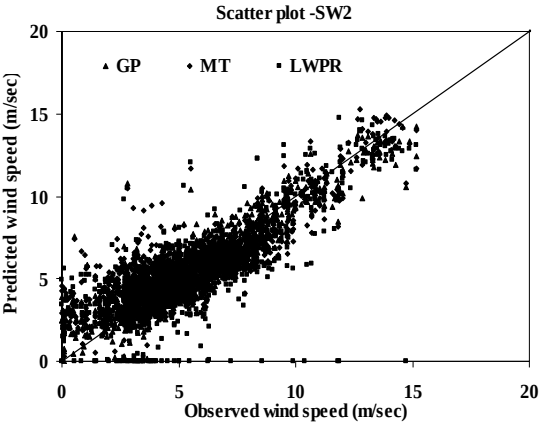
testing data to the duration: 15th January'06 to 17th July'06. Figure 4 (a) and (b) show the comparison between actual observations of the wind speed with corresponding model-evaluated values during testing. Quantitatively the testing errors were assessed through R, RMSE and MAE and these are shown in Table 5. Figures 4 (a) and (b) indicate fairly satisfactory performance by LWPR and also by the other methods. The level of accuracy obtained is noteworthy considering the highly varying and arbitrary nature of the wind. In general all the methods performed more or less equally. This is significant since SVR and LWPR are not purely non-linear like GP. It thus seems that the mathematical constraint may not always restrict accuracy of the model in general.

CONCLUDING REMARKS

The problem of hindcasting significant wave height and average wave period from continuous past measurements of wind speed and direction on the basis of the AI techniques of GP and MT was first discussed in this study. The data collected at an offshore location in the Arabian Sea were used. Accounting for wind speed and direction values observed over the preceding period of 24 hours was found to help the calibration process and yield better results, however comparable accuracy was still achieved when only current wind observations were used. Both the methods, GP and MT, performed satisfactorily in the given task of wind-wave simulation as reflected in high values of R, and low values of MAE, and RMSE. Although magnitudes of the error statistics did not indicate a significant difference between the performance of GP and MT, qualitative scatter diagrams and time histories showed the tendency of MT to estimate the higher waves more correctly. This is noteworthy since MT is not a purely non-linear model like GP and one would have expected GP to do well in all ranges of waves. However it appears that the splitting of input domain into sub-domains provides a special consideration of higher values and this helps in their better evaluation. The same conclusions as in the above case of significant wave heights can be drawn for the case of average wave period also. In general the estimation of 'Tz' was less satisfactory than that of 'Hs' and this can be expected in view of a highly varying nature of 'Tz' values.



(a)



(b)

Figure 4. Wind Speed prediction by GP, MT, LWPR

(a = time history; b = scatter plot)

The techniques of SVR and LWPR showed promising results in the application involving resolution of a property into constituents and derivation of an inverse causal relationship.

The applications of the techniques discussed in this paper may inspire similar studies in other tasks in coastal and ocean engineering and may help in producing highly efficient and flexible modeling procedures. It would be of interest to know how these computing tools work for different tasks such as developing a causal relationship, temporal regression, integration and resolution of a property. It is to be noted that while ANN and GP are purely non-linear soft computing tools, SVR, MT and LWPR are locally linear statistical techniques based on corresponding mathematical constraints. However equal performance of these methods seen in this study would indicate that in practice such constraints may not necessarily lower the performance of fitted model and hence advantage can always be taken of the transparent and portable solutions yielded by these schemes..

It is however felt that in order to derive maximum benefits of the data driven methods discussed in this paper a combination of these with physically based approaches may be attempted in future.

APPENDIX I. THE MODELING TOOLS USED

Genetic Programming

Genetic programming (Koza, 1992) involves considering given variables as terminals and mathematical, logical operators such as +, -, x, /, if-then-else, as functions. It forms innumerable combinations in between these functions and terminals. Every combination of individual is constructed by using genetic operators of reproduction, mutation, and cross over. Every time the relatively fit combination is asked to continue and others to vanish. The fittest combination decided as per some fitness criterion, such as the root mean square error between the observed and the estimated values, is the solution of the given problem.

Model Trees

Model tree (Witten and Frank, 2000) consists of dividing the entire input domain into sub-domains and thereafter fitting multiple linear regression models in each one of them. It therefore builds a piecewise linear model in order to approximate the given nonlinear relationship between a dependent variable and corresponding independent variables. For constructing model trees there are alternative algorithms and these depend on the domain splitting criterion used. The domain is split based on minimization of the entropy in sub-domains, or on collecting as many samples as possible belonging to the same class. Quinlan (1992)'s M5 algorithm is popular, in comparison with other options like regression splines. In the M5 type of algorithm the domain splitting is done with the help of standard deviation of the class value reaching a given class, which is a measure of the error at that class. The variable that maximizes such error reduction is selected for major or root splitting.

Support vector regression

In SVR we transform the input space into a feature space and fit linear models thereon using certain selected samples called support vectors. The transformation is achieved using Kernel functions of say Gaussian or polynomial type. Basically SVR optimally separates two or more classes in data by finding a demarcating hyperplane that maximizes the margin, or the highest distance between two closest points on both sides of the hyperplane, and also minimizes the error of misclassification. (Vapnik, 1997).

Locally weighted projection regression

LWPR consists of building linear regression models around given points of interest using data that are 'local' to such query points and kept in the memory. (Atkeson et al., 1997). In order to obtain output the local models fit data only in a selected region around the location of 'query' or new input. Locally weighted learning is memory based; explicitly retaining training data and every time using them for making a prediction. In locally weighted regression a surface rather than a line or curve is fitted to nearby points around the query point using a distance weighted regression. Locally weighted projection regression is a further division of the locally weighted regression and it employs models that are linear in selected (projected) directions of the input space. The input data are projected onto different orthogonal directions, along which a univariate linear regression is carried out and finally their average is taken to obtain the output for a given input.

Past works on GP, MT, SVR, and LWPR in ocean engineering

Applications of GP to carry out online prediction of waves can be seen in Jain and Deo (2008), and Gaur and Deo (2008). Kalra and Deo (2007) and Ustoorikar and Deo (2008) successfully applied GP to fill up gaps in wave data. The usefulness of GP in predicting coastal currents in real time can be seen in Charhate et al. (2007), while the same in estimating coastal sediment transport can be looked in Singh et al. (2007). Kalra et. al. (2008) presented how GP can be more effectively used than ANN in translating deep water waves to shallow water.

There are very few applications of MT reported so far in the field of ocean engineering and these incorporate the modeling of sediment transport in a harbor entrance channel (Bhattacharya et al., (2004) and Bhattacharya and Solomatine, (2006)).

The techniques of SVM or SVR have been also applied by some investigators in water resources (Dibike et. al, (2001) and An et. al. (2007)), but hardly any in the oceans. There is however a recent publication in the latter by Rajasekaran et al. (2008) in which authors attempted to predict storm surge with SVR.

The use of LWPR in ocean engineering so far again is almost nil except the one by Agarwal et al. (2008), who successfully applied it to estimate the coastal sediment transport as well as scour downstream a flip bucket spillway.

REFERENCES

- Agarwal M, Goyal M and Deo M C (2008): "Locally weighted polynomial regression for predicting hydraulic parameters", accepted in Civil Engineering and Environmental Systems, Taylor and Francis, DOI: 10.1080/10286600802517491 (<http://www.informaworld.com>)
- An S.H., Park U.Y., Kang K.I., Cho M.Y., Cho H.H. (2007): "Application of support vector machine in assessing conceptual cost estimates", Journal of computing in civil engineering, ASCE, 2007, Vol. 21, No. 4, pp. 259-264.
- Atkeson, C., Moore, A., and Schaal, S. (1997): "Locally weighted learning", Artificial Intelligence Review, 11 (4), 76–113.
- Bhattacharya B. and Solomatine D.P. (2006): "Machine Learning in sedimentation modeling". Neural Networks, Vol. 19, pp. 208-214.
- Bhattacharya P., Van Geer F., Van Der Veer P., Cser J. (2004): "Neural networks and model trees in spatio-temporal modeling of ground water head", 6th International Conference on Hydroinformatics, Liong et al. (eds), World Scientific, 1303-1310.
- Charhate S.B., M.C. Deo and Sanil Kumar V. (2007): "Soft and hard computing approaches for real time prediction of currents in a tide dominated area", Journal of Engineering for the Maritime Environment, Proceedings of the Institution of Mechanical Engineers, London, Part M, 221/2007, pp. 147-163.
- Coastal Engineering Manual (2003). Meteorology and Wave Climate. Engineer Manual; EM 1110-2-1100, U.S. Army Corps of Engineers, Washington, DC.
- Deo, M. C., Jha, A., Chaphekar, A. S., Ravikant, K. (2001). "Wave prediction using neural networks." Ocean Engineering, Elsevier, Oxford, U.K., 28(2001), 889-898.
- Dibike Y.B., Velickov S., Solomatine D. and Abbott M.B. (2001): "Model induction with support vector machines: Introduction and applications", Journal of computing in civil engineering, ASCE, Vol. 15, No. 3, pp. 208-216.
- Gaur S. and M C Deo (2008): "Real time wave forecasting using Genetic programming", Ocean Engineering, Elsevier, Vol. 35, pp. 1166-1172. http://www.ndbc.noaa.gov/station_page.php?station=44008
- Jain P. and M.C. Deo (2008): "Artificial intelligence tools to forecast ocean waves in real time", The Open Ocean Engineering Journal, Bentham Science, Vol. 1, pp. 13-21.
- Jain, P., and Deo, M. C.(2006)."Neural networks in ocean engineering." International Journal of Ships and Offshore Structures, Woodhead publications, Cambridge, UK, 1(1), 25-35.

- Kalra R. and M.C. Deo (2007): "Genetic Programming to retrieve missing information in wave records along the west coast of India", *Applied Ocean Research*, Elsevier, Vol. 29, No. 3, pp. 99-111.
- Kalra R. and M.C. Deo, Raj Kumar, Vijay K Agarwal (2008): "Genetic programming to estimate coastal waves from deep water measurements", *International Journal of Ecology and Development*, Indian Society for Development and Environment Research Roorkee, India, ISSN 0972-9984, Vol. 10, No. S08, pp. 67-76
- Koza, J. R. (1992). "Genetic Programming: On the Programming of Computers by Means of Natural Selection" The MIT press, 1992.
- Naithani, R., and Deo, M.C. (2005). "Estimation of wave spectral shapes using ANN", *Advances in Engineering Software*, Elsevier, 36(2005), 750-756.
- Namekar, S., and Deo, M. C. (2006). "Application of artificial neural network model in estimation of wave spectra", *ASCE Journal of Waterways, Port, Coastal and Ocean Division*, 132(5), 1-4.
- Quinlan, J.R. (1992). "Learning with continuous classes", *Proc. of Australian Joint Conf. on AI*, 343-348, World Scientific, Singapore.
- Rajasekaran S, Gayathri S and T L Lee (2008): "Support vector regression methodology for storm surge predictions", *Ocean Engineering*, 35(2008), 1578-1587.
- Sarpkaya, T. and Isaacson. M. (1981). "Mechanics of Wave Forces on Offshore Structures", Van Nostrand Reinhold, New York.
- Singh A.K., M.C. Deo and V. Sanil Kumar (2007): "Neural network – genetic programming for sediment transport, *Journal of Maritime Engineering*", The Institution of Civil Engineers, London, Vol. 160, No. MA3, pp. 113-119.
- Toussaint, M. (2006). "Locally Weighted Projection Regression Software", (<http://www.marc-toussaint.net/source-code/index.html>)
- Ustoorikar K. and M C Deo (2008): "Filling up Gaps in Wave Data with Genetic Programming", *Marine Structures*, Elsevier, Vol. 21, pp. 177-195.
- Vapnik, V. (1997). "Statistical Learning Theory". New York: John Wiley and Sons.
- Witten, I. H, Frank, E. (2000). "Data Mining: Practical machine learning tools and techniques", Morgan Kaufmann, San Francisco, 2000.
- Zamani A., Solomatine D., Azimian A., and Heemink A. (2008). "Learning from data for wind wave forecasting" *Journal of Ocean Engineering*, 35 (2008) 953-962